

Sahar Abdelnabi

AI/ML Security and Safety Researcher

✉ sahar.abdelnabi@tue.ellis.eu

🏠 <https://s-abdelnabi.github.io/>

🔗 Google Scholar 🌐 GitHub

RESEARCH FOCUS

I am interested in the broad intersection of AI, AGI, and multi-agent systems with security, safety, and sociopolitical aspects:

- 1) Understanding, probing, and evaluating the failure modes of AI models, their biases, emergent risks, and misuse scenarios.
- 2) Designing mitigations, system defenses, white-box control methods, reasoning enhancements, and novel architectures to counter such risks.
- 3) Leveraging AI agents for good: scientific discovery and advancing our society.

Our work in 2023 was the first to identify the indirect prompt injection vulnerability in LLM-integrated applications, and in 2020 the first to propose and call for watermarking generative AI. Our work on LLM sampling heuristics received a Best Paper Award at ACL 2025.

CURRENT POSITION

Principal Investigator and Independent Research Group Leader

Oct. 2025 – present

- ELLIS Institute Tübingen, Max Planck Institute for Intelligent Systems, and Tübingen AI Center, Germany
- I lead the COMPASS (COoperative Machine intelligence for People-Aligned Safe Systems) research group, developing safe, aligned, and steerable AI agents with an emphasis on security, safety, human aspects, and cooperative multi-agent systems.
 - ELLIS member; faculty member of the International Max Planck Research School for Intelligent Systems (IMPRS-IS); associated faculty member of the Max Planck ETH Center for Learning Systems (CLS).

PREVIOUS EXPERIENCE

AI Security Researcher (full-time)

Feb. 2024 – Sept. 2025

- Microsoft Security Response Center (MSRC), Microsoft Research Cambridge, UK
- Conducted AI security and safety research.
 - Assessed AI security vulnerabilities reported through Microsoft's AI Bug Bounty program.

Research Assistant

2017 – 2019

Max Planck Institute for Informatics, Germany. Advised by Prof. Dr. Andreas Bulling

Quality Assurance Engineer

2013 – 2017

Mentor Graphics (now Siemens EDA), Egypt

EDUCATION

Ph.D. in Computer Science

2019 – 2024

CISPA Helmholtz Center for Information Security & Saarland University, Germany
Advisor: Prof. Dr. Mario Fritz
Grade: **Summa cum laude** (External reviewers: Prof. Dr. Battista Biggio and Prof. Dr. Florian Tramèr)

M.Sc. in Computer Science

2017 – 2019

Saarland University, Germany. GPA: 1.2/1.0 (Thesis: 1.0). Advisor: Prof. Dr. Mario Fritz

M.Sc. in Computer and Systems Engineering

2013 – 2017

Ain Shams University, Egypt. GPA: 3.7/4.0

B.Sc. in Computer and Systems Engineering

2008 – 2013

Ain Shams University, Egypt

GRANTS AND FUNDING

- 2026 **Lead PI**, UK AI Security Institute [grant](#) on evaluation awareness (£662,715)
- 2026 **PI**, Coefficient Giving, project on AI for sound reasoning (sycophancy and decision making) (\$196,000)
- Co-PI**, Open Philanthropy grant on deception in LLMs (main PI: Ruta Binkyte) (\$88,550)
- 2023 Academic Research Grant for Google Cloud research credits
- 2021 Academic Research Grant for Google Cloud research credits

AWARDS AND HONORS

- 2026 Selected presentation, [KAUST Rising Stars in AI Symposium](#)
- 2025 **Best Paper Award**, ACL 2025 (awarded to 4 of more than 3,000 accepted papers), with oral presentation and panel discussion
- 2023 [Vector Distinguished Postdoctoral Fellowship](#) (declined)
- 2023 **Best Paper Award** and oral presentation, AISec 2023
- 2021 Oral presentation, ICCV 2021
- 2019 Saarland University scholarship for international students (DAAD STIBET III scholarship grant)
- 2016 IEEE Computational Intelligence Society (CIS) Outstanding Student-Paper Travel Grant, CIBCB 2016

PUBLICATIONS

* denotes equal contribution.

Conference Papers

- [C1] **Sahar Abdelnabi** and Eugene Bagdasarian. “AI Agents May Always Fall for Prompt Injections”. In: *NeurIPS*. 2026.
- [C2] Katharina Deckenbach*, Haritz Puerto*, Jonas Geiping, and **Sahar Abdelnabi**. “Models That Know How Evaluations Are Designed Score Safer”. In: *NeurIPS Evaluations & Datasets*. 2026.
- [C3] Mason Nakamura*, Abhinav Kumar*, Saswat Das*, **Sahar Abdelnabi**, Saaduddin Mahmud, Ferdinando Fioretto, Shlomo Zilberstein, and Eugene Bagdasarian. “Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems”. In: *NeurIPS Evaluations & Datasets*. 2026.
- [C4] Sina Mavali, David Pape, Jonathan Evertz, Samira Abedini, Devansh Srivastav, Thorsten Eisenhofer, **Sahar Abdelnabi**, and Lea Schönherr. “No More, No Less: Task Alignment in Terminal Agents”. In: *NeurIPS Evaluations & Datasets*. 2026.
- [C5] David Schmotz, Luca Beurer-Kellner, **Sahar Abdelnabi**, and Maksym Andriushchenko. “Skill-Inject: Measuring Agent Vulnerability to Skill File Attacks”. In: *NeurIPS Evaluations & Datasets*. 2026.
- [C6] **Sahar Abdelnabi**, Aideen Fay, Ahmed Salem, Egor Zverev, Kai-Chieh Liao, Chi-Huang Liu, Chun-Chih Kuo, Jannis Weigend, Danyael Manlangit, Alex Apostolov, et al. “LLMail-Inject: A Dataset from a Realistic Adaptive Prompt Injection Challenge”. In: *EMNLP Findings*. 2026.
- [C7] Ivaxi Sheth, Jan Wehner*, **Sahar Abdelnabi***, Ruta Binkyte*, and Mario Fritz. “Position: Safety Must Precede the Deployment of Open-Ended AI Agents”. In: *ICML*. 2026.
- [C8] Ahmed Salem, Andrew Paverd, and **Sahar Abdelnabi**. “Position: Stateless Yet Not Forgetful: Implicit Memory as a Hidden Channel in LLMs”. In: *SaTML*. 2026.
- [C9] Amr Gomaa, Ahmed Salem, and **Sahar Abdelnabi**. “ConVerse: Benchmarking Contextual Safety in Agent-to-Agent Conversations”. In: *EACL Findings*. 2026.
- [C10] **Sahar Abdelnabi** and Ahmed Salem. “The Hawthorne Effect in Reasoning Models: Evaluating and Steering Test Awareness”. In: *NeurIPS. Spotlight*. 2025.
- [C11] Guangchen Lan, Huseyin A. Inan, **Sahar Abdelnabi**, Janardhan Kulkarni, Lukas Wutschitz, Reza Shokri, Christopher G. Brinton, and Robert Sim. “Contextual Integrity in LLMs via Reasoning and Reinforcement Learning”. In: *NeurIPS*. 2025.
- [C12] Ivaxi Sheth, **Sahar Abdelnabi**, and Mario Fritz. “Context-Aware Reasoning On Parametric Knowledge for Inferring Causal Variables”. In: *EMNLP Findings*. 2025.
- [C13] Sarath Sivaprasad*, Pramod Kaushik*, **Sahar Abdelnabi**, and Mario Fritz. “A Theory of Response Sampling in LLMs: Part Descriptive and Part Prescriptive”. In: *ACL (main). Oral & Panel Discussion & Best Paper Award*. 2025.
- [C14] **Sahar Abdelnabi***, Aideen Fay*, Giovanni Cherubin, Ahmed Salem, Mario Fritz, and Andrew Paverd. “Get My Drift? Catching LLM Task Drift with Activation Deltas”. In: *SaTML*. 2025.
- [C15] Egor Zverev, **Sahar Abdelnabi**, Soroush Tabesh, Mario Fritz, and Christoph H. Lampert. “Can LLMs Separate Instructions From Data? And What Do We Even Mean By That?” In: *ICLR*. 2025.
- [C16] **Sahar Abdelnabi**, Amr Gomaa, Sarath Sivaprasad, Lea Schönherr, and Mario Fritz. “Cooperation, Competition, and Maliciousness: LLM-Stakeholders Interactive Negotiation”. In: *NeurIPS Datasets and Benchmarks*. 2024.

- [C17] Edoardo Debenedetti*, Javier Rando*, Daniel Paleka*, Silaghi Fineas Florin, Dragos Albastroiu, Niv Cohen, Yuval Lemberg, Reshmi Ghosh, Rui Wen, Ahmed Salem, Giovanni Cherubin, Santiago Zanella-Beguelin, Robin Schmid, Victor Klemm, Takahiro Miki, Chenhao Li, Stefan Kraft, Mario Fritz, Florian Tramèr, **Sahar Abdelnabi**, and Lea Schönherr. “Dataset and Lessons Learned from the 2024 SaTML LLM Capture-the-Flag Competition”. In: *NeurIPS Datasets and Benchmarks. Spotlight*. 2024.
- [C18] **Sahar Abdelnabi** and Mario Fritz. “Fact-Saboteurs: A Taxonomy of Evidence Manipulation Attacks against Fact-Verification Systems”. In: *USENIX Security*. 2023.
- [C19] Giada Stivala, **Sahar Abdelnabi**, Andrea Mengascini, Mariano Graziano, Mario Fritz, and Giancarlo Pellegrino. “From Attachments to SEO: Click Here to Learn More about Clickbait PDFs!” In: *Annual Computer Security Applications Conference (ACSAC)*. 2023.
- [C20] **Sahar Abdelnabi**, Rakibul Hasan, and Mario Fritz. “Open-Domain, Content-based, Multi-modal Fact-checking of Out-of-Context Images via Online Resources”. In: *CVPR*. 2022.
- [C21] **Sahar Abdelnabi** and Mario Fritz. “Adversarial Watermarking Transformer: Towards Tracing Text Provenance with Data Hiding”. In: *S&P*. 2021.
- [C22] Ning Yu*, Vladislav Skripniuk*, **Sahar Abdelnabi**, and Mario Fritz. “Artificial Fingerprinting for Generative Models: Rooting Deepfake Attribution in Training Data”. In: *ICCV. Oral Presentation*. 2021.
- [C23] **Sahar Abdelnabi**, Katharina Krombholz, and Mario Fritz. “VisualPhishNet: Zero-Day Phishing Website Detection by Visual Similarity”. In: *CCS*. 2020.
- [C24] **Sahar Abdelnabi**, Michael Xuelin Huang, and Andreas Bulling. “Towards High-Frequency SSVEP-Based Target Discrimination with an Extended Alphanumeric Keyboard”. In: *IEEE International Conference on Systems, Man and Cybernetics (SMC)*. 2019.
- [C25] **Sahar Abdelnabi**, Seif Eldawlatly, and Mahmoud I. Khalil. “Epileptic Seizure Prediction Using Zero-Crossings Analysis of EEG Wavelet Detail Coefficients”. In: *IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB)*. 2016.

Journal Articles

- [J1] **Sahar Abdelnabi***, Amr Gomaa*, Eugene Bagdasarian, Per Ola Kristensson, and Reza Shokri. “Firewalls to Secure Dynamic LLM Agentic Networks”. In: *Transactions on Machine Learning Research (TMLR)* (2026).
- [J2] Jan Wehner, **Sahar Abdelnabi**, Daniel Tan, David Krueger, and Mario Fritz. “Taxonomy, Opportunities, and Challenges of Representation Engineering for Large Language Models”. In: *Transactions on Machine Learning Research (TMLR)* (2025). **Survey Certification**.

Workshop Papers

- [W1] Rebecca Weil, **Sahar Abdelnabi**, Mario Fritz, and Rakibul Hasan. “Tell Me What You Like and I Know What You Will Share: Topical Interest Influences Behavior Toward News from High and Low Credible Sources”. In: *IEEE EuroS&P Workshops*. 2024.
- [W2] Kai Greshake*, **Sahar Abdelnabi***, Shailesh Mishra, Christoph Endres, Thorsten Holz, and Mario Fritz. “Not What You’ve Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection”. In: *AISeC Workshop, in conjunction with CCS. Oral Presentation, Best Paper Award*. 2023.
- [W3] **Sahar Abdelnabi** and Mario Fritz. “What’s in the Box: Deflecting Adversarial Attacks by Randomly Deploying Adversarially-Disjoint Models”. In: *The 8th ACM Workshop on Moving Target Defense (MTD), in conjunction with CCS*. 2021.

Preprints

- [P1] **Sahar Abdelnabi**, Chris Hicks, Konrad Rieck, and Ahmad-Reza Sadeghi. “Measuring Security Without Fooling Ourselves: Why Benchmarking Agents Is Hard”. arXiv:2605.22568. 2026.
- [P2] Changling Li, Terry Jingchen Zhang, Jie Zhang, Zhijing Jin, **Sahar Abdelnabi**, and Maksym Andriushchenko. “Decomposing and Measuring Evaluation Awareness”. arXiv:2605.23055. 2026.
- [P3] Sidharth Pulipaka, Stanislau Hlebig, Leonidas Raghav, **Sahar Abdelnabi**, Vyas Raina, Ivaxi Sheth, and Mario Fritz. “Hidden in Memory: Sleeper Memory Poisoning in LLM Agents”. arXiv:2605.15338. 2026.
- [P4] Aaron Rose, Carissa Cullen, **Sahar Abdelnabi**, Philip Torr, Brandon Gary Kaplowitz, and Christian Schroeder de Witt. “Detecting Multi-Agent Collusion Through Multi-Agent Interpretability”. arXiv:2604.01151. 2026.
- [P5] Mason Nakamura*, Abhinav Kumar*, Saaduddin Mahmud, **Sahar Abdelnabi**, Shlomo Zilberstein, and Eugene Bagdasarian. “Terrarium: Revisiting the Blackboard for Multi-Agent Safety, Privacy, and Security Studies”. arXiv:2510.14312. 2025.
- [P6] David Schmotz, **Sahar Abdelnabi**, and Maksym Andriushchenko. “Agent Skills Enable a New Class of Realistic and Trivially Simple Prompt Injections”. arXiv:2510.26328. 2025.

TALKS AND PANELS

Invited Talks

AI Agents May Always Fall for Prompt Injections: Casting Prompt Injection via the Lens of Contextual Integrity
KU Leuven Summer School on Security and Privacy in the Age of AI, 2026

Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems
Updates in Cooperative AI seminar series, 2026

The Challenge of Test Awareness in AI Evaluation (selected talk)
KAUST Rising Stars in AI Symposium, 2026

What Does It Mean for AI Agents to Preserve Privacy?
Huawei Trustworthy AI Summit, 2026; Graz Security Week (summer school), 2025

Firewalls to Secure Dynamic LLM Agentic Networks
Brave, 2025; Google DeepMind, 2025; Qualcomm, 2025

Towards Aligned, Interpretable, and Steerable Safe AI Agents
TU Graz, 2025; UMass Amherst Security and Privacy Seminar, 2025; CISPA, 2025; ELLIS Institute Tübingen Scientific Symposium, 2025

On the Security of Real-World LLM-Integrated Applications
European Symposium on Security and Artificial Intelligence (ESSAI), 2024

On New Security and Safety Challenges Posed by LLMs and How to Evaluate Them
Keynote, HIDA PhD Meet-up, 2024; MLSec Seminars, 2024

On Evaluating Language Models and Their Security and Safety Implications
Vector Institute, 2023; ETH Zürich, 2023

LLM-Deliberation: Evaluating LLMs with Interactive Multi-Agent Negotiation Games
ACL SIGSEC, 2023

Compromising LLMs: The Advent of AI Malware
Black Hat USA, 2023

Not What You've Signed Up For: Investigating the Security of LLM-Integrated Applications
Privacy and Security in ML Seminars, 2023

How to Improve Automated Fact-Checking
Max Planck Institute for Software Systems, 2022

Multi-modal Fact-checking: Out-of-Context Images and How to Catch Them
UCL Information Security Seminars, 2022

Panels

Science for AI Workshop: Shaping Public AI for Science, Innovation & European Impact
AI in Science Summit, 2025

Women in AI Security Workshop
The Alan Turing Institute, 2025

Security of Generative AI and Generative AI in Security
DIMVA, 2023

Research Cohort: Implementation and Evaluation of a Research Paper
AI Saturdays Lagos, 2024

TEACHING

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| 2026 | Lecturer , University of Tübingen. <i>AI Safety</i> (co-taught with Maksym Andriushchenko and Jonas Geiping) |
| 2020–23 | Teaching Assistant , Saarland University. “Opportunities and Risks of Large Language Models and Foundation Models” (seminar), “Machine Learning in Cybersecurity”, “High-level Computer Vision” |
| 2017–19 | Tutor , Saarland University. “Image Processing and Computer Vision”, “Neural Networks Implementation and Applications”, “Interactive Systems” |

SUPERVISION AND MENTORING

COMPASS group (current)

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|-----------------|---|
| <i>Postdoc</i> | Haritz Puerto |
| <i>PhD</i> | Daniel Donnelly; Jeanne Salle (co-advised with Maksym Andriushchenko); Katharina Deckenbach; Johannes Taraz; Evan Chapple |
| <i>Visiting</i> | Abhinav Kumar; Haneen Najjar (visiting PhD students) |
| <i>Alumni</i> | Luca Scionis (visiting PhD student) |

Fellowship mentoring

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| 2025 | Mentor, Cooperative AI Research Fellowship |
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ACADEMIC SERVICE

Policy and Advisory

- 2026– **Member of the EU AI Act Scientific Panel**, a panel of 60 independent experts advising the AI Office and national authorities on the implementation of the AI Act and the assessment of the impacts and risks of general-purpose AI (GPAI) models
Grant reviewer for the Cooperative AI Foundation

Workshop Organization

- 2026 Co-chair, AISec 2026
2026 Co-organizer, PALM workshop at NeurIPS 2026
2026 Co-organizer, 2nd Workshop on Epistemic Intelligence in Machine Learning at ICML 2026
2025 Co-organizer, Foundations of Language Model Security workshop at EurIPS 2025
2025 Co-organizer, LLM Safety and Security Workshop at the ELLIS UnConference

Competition Organization

- 2025 Lead organizer, IEEE SaTML 2025 challenge “LLMail-Inject: Adaptive Prompt Injection Attacks”
2024 Co-organizer, IEEE SaTML 2024 challenge “LLM Capture-the-Flag”

Program Committees

- AC Associate Chair, IEEE S&P 2027
PC IEEE S&P ('26), ACM CCS ('25, '26), USENIX Security ('25), IEEE SaTML ('24, '26), AAAI ('25), AISec Workshop ('23–'25)

Reviewing

ICML ('24, '26), ICLR ('24–'26), NeurIPS ('23, '25), CVPR ('22, '23), ICCV ('23), ECCV ('22), IEEE TPAMI (2021, 2022, 2024), TMLR
Workshops: ICML'23 Neural Conversational AI Workshop; ICLR'21 Workshop on “Synthetic Data Generation – Quality, Privacy, Bias”

OUTREACH AND MEDIA

Press and Interviews

- 2023 Our work on indirect prompt injection was featured with interviews in Vice, Wired, Zeit, MIT Technology Review, and CISPA communication channels

Podcasts and Documentaries

- 2023 *ChatGPT: What happens when the AI takes over?*, Y-Kollektiv documentary
2023 *A dark side to LLMs*, CyberWire podcast
2022 *Deepfakes and Fingerprinting*, CISPA tl;dr podcast

Blogs

- 2024–25 Microsoft Security Response Center (MSRC): [Announcing the Adaptive Prompt Injection Challenge \(LLMail-Inject\)](#) and [Announcing the winners of the Adaptive Prompt Injection Challenge \(LLMail-Inject\)](#)
Our work on LLM-deliberation was featured by the Montreal AI Ethics Institute

Policy and Industry Impact

Our work on indirect prompt injection has been featured by policymakers and practitioners such as the German Federal Office for Information Security, NIST, OWASP, MITRE, Microsoft's AI bug bar, and many others, and introduced new terminology for the entire research and tech fields

LANGUAGES

English (Fluent/Bilingual), German (Intermediate), Colloquial Egyptian (Mother Tongue)